535/3
PHYSICS
(Practical)
PAPER 3
JULY/AUGUST 2026
2 HOURS

RWENZORI REGION ASSESSMENT BOARD

UGANDA CERTIFICATE OF EDUCATION

PHYSICS

(PRACTICAL)

PAPER 3

2 Hours

INSTRUCTIONS TO THE CANDIDATES:

- This paper consists of **two** examination items.
- Answer only **one** item.
- Any additional items will **not** be scored.
- You are **not** allowed to start working with the apparatus for the first quarter of an hour. This time is to enable candidates: read the items thoroughly, check for the apparatus they need and plan appropriately.
- Answers to the item must be written in the answer booklet provided.
- Graph paper is provided.
- Mathematical tables and silent non-programmable calculators may be used.

Item 1: A camera man gave out his camera to his good friend to use it on his weeding taking photos but when his friend returned the camera, the man complained of his camera lens not focusing well the objects, he then decided to take the camera to the mechanic who advised him to replace the lens camera with a new one because it had been used for a long time. The lens in the man's camera was of a focal length ranging from **9cm to 15cm**. The man visited shop selling the camera lenses and presented the mechanic's recommendation and then bought the lens. The man however, was not sure about the focal of the lens he purchased from the shop because the problem persisted.

Task: You are provided with a lens, which has same properties with that of the camera lens the man bought. Verify the accuracy of a lens in the camera the man bought.

Hint: In your report include: the aim, variables (independent, dependent and controlled), hypothesis, procedure, possible sources of errors, precautions to the stated errors, data presentation (at least five data sets), data analysis using a graph and advise/recommendation.

Item 2: According to the information provided by Physics teacher to laboratory technician of a certain school, the technician was instructed to provide learners with a helical spring of force constant in the range of **$20Nm^{-1}$ to $35Nm^{-1}$**. But the technician without the knowledge of force constant gave the learners the springs she got from the laboratory. On reaching the physics laboratory for the lesson, the concerned teacher checked the springs and was doubtful of their force constant. The concerned teacher then picked one of the springs and gave it to one of the learners to verify the force constant.

Task: As a learner of physics, carry out a scientific investigation to help the concerned teacher determine the force constant of the springs provided.

Hint: In your report include: the aim, variables (independent, dependent and controlled), hypothesis, procedure, possible sources of errors, precautions to the stated errors, data presentation (at least five data sets), data analysis using a graph and advise/recommendation.

END:.....